

Week 1 Data Processing Notes

Main focus: clean the transaction table | Extension: create analysis-ready feature tables

Overview: This document briefly explains the current data-processing work. The main focus is to clean the original transaction table first. The four additional feature tables are prepared as a small extension for the next step of analysis.

1. Clean the Transaction Table First

What I did:

- Converted InvoiceDate into a standardized datetime field called OrderDate.
- Converted Quantity and UnitPrice into numeric fields so they can be used for calculation.
- Removed rows without CustomerID, because customer analysis needs to know who made the purchase.
- Removed rows where Quantity or UnitPrice is less than or equal to 0, because they are not normal sales records for this stage.
- Removed exact duplicate transaction rows to avoid double-counting sales.
- Created Revenue = Quantity * UnitPrice to represent the sales amount of each product line.
- Created OrderMonth to support monthly trend analysis later.
- Kept the standardized OrderDate in the final table and removed the original InvoiceDate string to avoid confusion later.

Cleaning result: The original dataset has 541,909 rows. After cleaning, the final transaction-level dataset has 392,692 rows and 10 columns. The cleaned table has no missing CustomerID, no abnormal quantity or price values, no exact duplicate rows, and the Revenue field has been validated.

2. Create Feature Tables from the Cleaned Product-Line Table

This part is a small extension, not the main task. The purpose is to prepare the cleaned data for the next step, such as EDA, dashboard building, customer analysis, product analysis, and country-level market analysis. The original cleaned_sales.csv file is not changed during this step.

- **order_features.csv:** one row per order. It shows how much each order is worth, how many items were bought, and how many product types were included.
- **customer_features.csv:** one row per customer. It shows how many orders each customer made, total spending, average order value, and how long the customer was active.
- **product_features.csv:** one row per product. It shows how many units each product sold, how much revenue it contributed, and how many customers bought it.
- **country_features.csv:** one row per country. It shows order count, customer count, total revenue, and average order value for each country.

3. Notes on the Code Folder

The whole Code folder can be zipped and submitted directly. The notebooks now look for data inside the dataset folder under the same Code directory, so they do not depend on an absolute path on my own computer.

File / Folder	Purpose
1.1_Preprocess.ipynb	Main notebook. It reads data.csv, checks data quality, cleans the dataset, and exports cleaned_sales.csv.
1.2_Feature_Aggregation.ipynb	Extension notebook. It reads cleaned_sales.csv and creates order, customer, product, and country feature tables.
dataset/data.csv	Original transaction-detail dataset.
dataset/cleaned_sales.csv	Cleaned transaction-level dataset, used as the base for later analysis.
dataset/feature_tables/	Four aggregated feature tables for later analysis from different business views.
docs/	Explanation document and the Week 1 task PDF if included.

4. Short Update for Manager

Suggested wording: I cleaned the transaction-level dataset by standardizing date and numeric fields, removing invalid or incomplete sales records, dropping exact duplicates, and creating Revenue and OrderMonth for later analysis. As an extension, I also created order-level, customer-level, product-level, and country-level feature tables to support the next phase of EDA and dashboard reporting.